ARTEFACT LATAM - WRITTEN TECHNICAL CASE SUBMISSION

AI Engineer - Full-Stack para Aplicações com IA Generativa

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GitHub Repository: github.com/IgorSouza-ui/artefact-ai-emporio | Greenhouse Application ID: 304279160002

1. Executive Summary & Challenge Scenario

This project implements a production-grade, highly resilient AI customer support agent in Python for Empório da Música, a traditional musical instruments retailer located in Campo Grande, Mato Grosso do Sul (founded in 2008). The system resolves customer service overload across text messaging channels by providing accurate, empathetic, and transparent assistance regarding product catalog discovery, live stock verification, postal order tracking, and store policies under the Brazilian Consumer Protection Code (CDC - Law 8,078/90).

2. Technical Architecture & Key Decisions

• Framework & Pattern: Native Python Function Calling / ReAct loop paired with lightweight RAG. Rather than using opaque third-party frameworks that introduce latency and runtime fragility, the solution maintains clean separation of concerns: an LLM reasoning engine, a structured SQLite query layer, and a curated policy retrieval index.
• Multi-Provider LLM Orchestration: Built with native support for Google Gemini (gemini-2.5-flash) via the official google-genai SDK, with pluggable support for OpenAI (gpt-4o-mini). Crucially, the system includes an Intelligent Local Engine Fallback with stateful multi-turn context memory, allowing evaluators to run the interactive CLI immediately.
• Structured Data Store (SQLite): Ingests all 6 CSV operational tables (categories, products, orders, order items, customers, promotions) into an indexed SQLite engine (emporio.db), enabling sub-millisecond filtering on price, stock quantities, and tracking codes.
• Policy Retrieval Engine (RAG): Indexes the 8-page políticas_da_loja.pdf manual into 9 functional domain sections (store hours, address, refund regret, warranties, payment discounts, LGPD) utilizing a BM25-inspired keyword and intent-boosted retriever.

3. Business Logic, Guardrails & Edge Cases

• Strict Real-Time Stock Enforcement: Product queries enforce stock_quantity > 0. Out-of-stock items (such as the Giannini GF-3D with 0 stock) are never presented as available, preventing false customer commitments.
• Zero Hallucinated Discounts: In promotions.csv, promotions carry an is_active boolean. When an expired campaign is queried (e.g., the 12% discount on Takamine GD20 where is_active = 0), the agent transparently clarifies that the promotion expired and presents the valid table price (R$ 2.199,00) alongside lawful payment terms (5% PIX discount = R$ 2.089,05; 12x credit card).
• CDC Compliance: Strictly applies Article 49 of the Brazilian Consumer Defense Code: 7 calendar days regret window for remote/online purchases, 100% store-covered return shipping, and 10-business-day refund.
• Strict Scope Boundaries: The store exclusively sells complete musical instruments. Inquiries regarding loose accessories (strings, picks, cables, pedals, amps) or home appliances are politely and clearly redirected.

4. Sample Interaction Transcripts Summary

Five comprehensive transcripts covering diverse scenarios were generated and included in docs/conversations/:

File	Scenario & User Query	Tools Invoked	Observed Agent Behavior
conversation_01	Acoustic Guitars (Violões) under R$ 1,000 in stock	search_catalog	Returned 5 guitars in stock; excluded zero-stock models; calculated PIX discounts & installments.
conversation_02	Store address & Saturday hours	search_store_policies	Provided Rua 14 de Maio 3200, phone (67) 3341-4444, and Saturday schedule (09:00 - 13:00).
conversation_03	Takamine GD20 & 12% discount query	get_product_info	Detected expired promotion (is_active=0); refused false discount; gave table price R$ 2.199,00 and stock (5).
conversation_04	Customer regret on online order	search_store_policies	Applied CDC Art. 49: 7 days window, store-paid return freight, 10-day refund timeframe.
conversation_05	Order tracking BRJL5544332BR + accessories inquiry	check_order_status search_store_policies	Found live order in transit (Kala Ukulele, delivery 2026-02-28); politely rejected out-of-scope items.

5. Automated Test Suite & Validation

A rigorous automated test suite in tests/test_emporio.py verifies all functional requirements. All 15 tests pass with zero regressions (python -m pytest tests/ -v in 0.76s):

- 5 catalog & data integrity tests (price filters, zero-stock exclusions, expired promotion rejection, active promotion calculations, order tracking).
- 4 policy retrieval tests (address resolution, Saturday hours, CDC 7-day regret, 5% PIX discount).
- 6 conversational agent guardrail tests (catalog response formatting, address guidance, expired promo handling, CDC refund terms, out-of-scope rejection, and multi-turn technical spec refinements like steel string 43mm nut under R$ 500).

6. AI Assistant Usage & Engineering Workflow

In accordance with Artefact's instructions to disclose AI assistant usage: Antigravity (powered by Google Gemini) was used as an agentic pair-programming assistant. It accelerated initial dataset parsing, generated comprehensive pytest assertions, and assisted with Markdown formatting. All architectural patterns (SQLite schema indexing, deterministic fallback engine, prompt guardrails, and CDC legal compliance) were designed, directed, and rigorously verified by the candidate.

Public Repository: https://github.com/IgorSouza-ui/artefact-ai-emporio
Delivered on October 31, 2026. Ready for live evaluation via python main.py.